Jean-Guillaume de Damas

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Experience

CERMICS, CNRS, ENPC, Institut Polytechnique de Paris, Postdoctoral Researcher — Marne-la-Vallée, France
May 2026 – present
1 month

- Working at the Centre d'Enseignement et de Recherche en Mathématiques et Calcul Scientifique on the development of hybrid model reduction methods with Virginie Ehrlacher
- Developing numerical methods for accelerating parametric studies in granular material simulations
- Implementing algorithms in the software SCOPI++

CERFACS, Master Intern - Researcher & Scientific Programmer — Toulouse, France
Apr 2022 – Oct 2022
7 months

- Researched randomized SVD preconditioners with applications to data assimilation
- Delivered functional C++ code for the European Weather Forecast Center
- Supervised by Selime Gürol

Education

PhD **INRIA Paris & Sorbonne Université**, Applied Mathematics — Paris, France
Nov 2022 – Mar 2026

- Thesis: Randomized methods for solving eigenvalue problems and computing low-rank matrix approximations
- Obtained up to 3-4x speed-up compared to state of the art methods while maintaining accuracy in the computations
- Supervised by Laura Grigori within the ERC project EMC2

MSc **ISAE-SUPAERO**, Engineering — Toulouse, France
Sept 2018 – Nov 2022

- Machine learning: Algorithms for SVM, Neural Networks, Reinforcement learning and mathematical background
- Statistics: Bayesian inference, Stochastic algorithms, Non-parametric estimation
- High performance computing: Parallelization in C
- Optimisation: Inverse problem, data assimilation, problem regularization, and numerical challenges in inexact arithmetic

Publications

Randomized Implicitly Restarted Arnoldi Method for the Non-Symmetric Eigenvalue Problem — Oct 2025
Vol. 46, No. 4, pp. 2395–2422. ISSN: 1095-7162.
Jean-Guillaume de Damas, Laura Grigori
10.1137/24m1674352 (SIAM Journal on Matrix Analysis and Applications)

Randomized Krylov-Schur Eigensolver with Deflation — Aug 2025
Jean-Guillaume de Damas, Laura Grigori
10.48550/ARXIV.2508.05400 (arXiv)

Randomized Orthogonalization and Krylov Subspace Methods: Principles and Algorithms — Nov 2025
Jean-Guillaume de Damas, Laura Grigori, Igor Simunec, Edouard Timsit
10.48550/arXiv.2512.15455 (arXiv)

Selected Talks

Workshop on Approximate Computing in Numerical Linear Algebra
A randomized Krylov-Schur eigensolver, featuring mixed-precision and deflation
Oct 2025

CERMICS Applied Mathematics Seminar
Randomized methods for the eigenvalue problem and low rank approximation
June 2025

SIAM Conference on Applied Linear Algebra LA24 & CERFACS Sparse Days
Randomized Implicitly Restarted Arnoldi algorithm for the unsymmetric eigenvalue problem
May 2024

Teaching

Mathematics for Science Studies I (52h)
Sorbonne University, Paris
Fall 2023

Programming Basics I (20h)
Sorbonne University, Paris
Fall 2023

Skills

Programming
Julia, Python, Matlab, R, C, C++

Technologies
PyTorch, Scikit-learn, OpenMP, MPI, Cuda

Expertise
Numerical Analysis, Randomized Linear Algebra, Eigensolvers, Low-Rank Approximations

Languages

English
Advanced Level C1 (TOEFL ITP 600/677)

German
Intermediate (A2)

Japanese
Beginner (A1)

Outside of the office

- Piano & Musicology: Learning as a passionate amateur with conservatory-level classes
- Game developement: Building its own engine in C++ or using Unity/Unreal